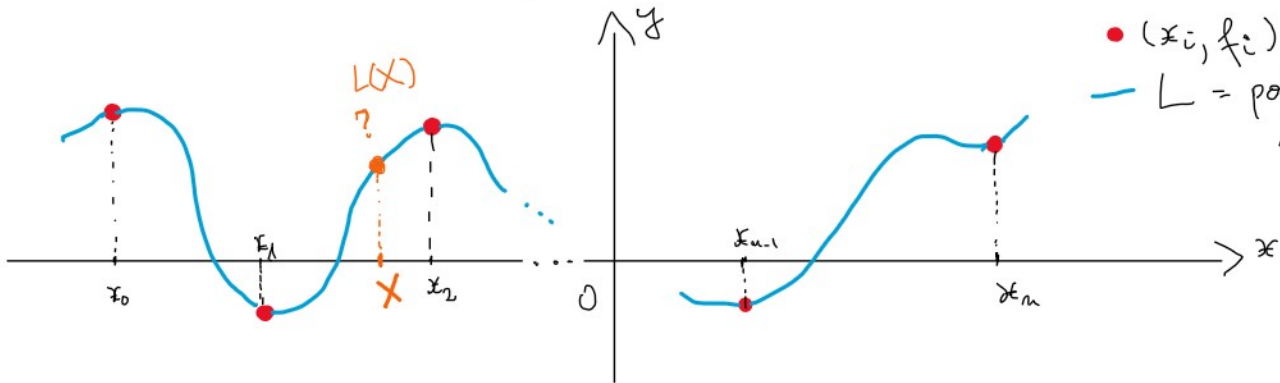


# Interpolare Lagrange

(„trece prin puncte“)



•  $(x_i, f_i)$   
 —  $L$  = pol. de interpolare de grad min.

Input : • nodes:  $x_0, x_1, \dots, x_m \rightarrow$  distincte  
 • values:  $f_0, f_1, \dots, f_m \rightarrow$  nu neapărat dist.  
 • pol(x):  $X$

Output:  $L(X)$

$L(x_i) = f_i$ ,  $i = \overline{0, m}$  (\*)  $\rightarrow$  interpolare

$$L(X) = a_0 + a_1 X + \dots + a_m X^m$$

$$(*) \Leftrightarrow \underbrace{\begin{bmatrix} 1 & x_0 & \dots & x_0^m \\ 1 & x_1 & \dots & x_1^m \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_m & \dots & x_m^m \end{bmatrix}}_{\text{Vandermonde}} \cdot \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_m \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ \vdots \\ f_m \end{bmatrix}$$

grad  $L \rightarrow$  min.  
 $\leq m = \text{nr. de noduri} - 1$

Alternativă: formula baricentrică

